$^{40}AR/^{39}AR$ AGE CALIBRATION OF THE LITHO- AND PALEOMAGNETIC STRATIGRAPHIES OF THE NGORORA FORMATION, KENYA[1]

ALAN DEINO, LISA TAUXE[2], MARC MONAGHAN[3], AND ROBERT DRAKE
Geochronology Center of the Institute of Human Origins, 2453 Ridge Road, Berkeley, California 94709

ABSTRACT

Precise eruptive ages have been determined by the laser-fusion, single-crystal $^{40}Ar/^{39}Ar$ method for juvenile volcanic feldspars from reworked and contaminated volcaniclastic rocks of the middle Miocene Ngorora Formation, Kenya Rift Valley. These ages range from 13.06 Ma at the base to 10.51 Ma toward the top of the type section near Kabarsero. Correlation of the local paleomagnetic stratigraphies with the geomagnetic reversal time scale yields magnetochronologic age estimates that are younger than the isotopic ages by an average of 0.18 Ma. Much of the discrepancy can be eliminated if an inferred change in sea-floor spreading rate occurred at 13 Ma or earlier, rather than at 10.42 Ma as previously suggested. Sedimentation rates at Kabarsero calculated from the $^{40}Ar/^{39}Ar$ results decrease from initial values of ~25 cm/1000 yr to ~5 cm/1000 yr toward the top of the section. The initial rapid sedimentation rates characterize the first 0.1 to 0.3 m.y. following emplacement of the underlying, voluminous, basin-filling Tiim Phonolites, indicating that the Baringo Basin at this time may not have existed as a rift valley created by extensional tectonics, but instead may have been a subsidence feature formed in response to removal of large volumes of magma from the lithosphere. A premolar tentatively identified as *Proconsul* sp. indet. found in the Ngorora Formation near the village of Bartabwa has been dated at ~12.42 Ma, representing perhaps the last known occurrence of this genus in the fossil record.

INTRODUCTION

Determining the depositional age of sedimentary strata is often a particularly vexing task. One often finds that even after establishing a biostratigraphy and a paleomagnetic stratigraphy calibrated by a number of K-Ar dates, one is unable to assign an age to an individual stratum or to a sequence of strata with a great deal of confidence. This is due, in part, to the ambiguity inherent in paleomagnetic and biostratigraphic correlations, and to the scarcity of uncontaminated volcanic strata in most sedimentary basins. Recent advances achieved by single-crystal $^{40}Ar/^{39}Ar$ dating promise to help resolve these difficulties by making it possible to determine accurate and precise eruption ages for even heavily contaminated volcaniclastic deposits (e.g., York et al. 1981; Lo Bello et al. 1987; Drake et al. 1988; Deino and Potts 1990). We have applied this technique to volcaniclastic strata from three sections of the Ngorora Formation, Baringo District, Kenya, with several goals in mind: (1) to assist in correlating the local paleomagnetic stratigraphy to the Geomagnetic Reversal Time Scale (GRTS); (2) to constrain reversal ages in the middle Miocene portion of the GRTS; (3) to calibrate the Ngorora Formation biostratigraphy, in particular the occurrence of hominoids, and (4) to provide age calibration for the lithostratigraphy of the formation and its bounding lava sequences to understand better the development of the central Kenya Rift during the middle Miocene.

The Ngorora Formation of the Kenya Rift Valley (fig. 1) is one of the most important sources of Miocene plant and vertebrate fossils, including hominoids, from sub-Saharan Africa (Hill et al. 1985; Hill and Ward 1988; and references therein). Tauxe et al. (1985) presented a chronostratigraphy for the type section of the Ngorora Formation at Kabarsero based on a paleomagnetic stratigraphy and conventional K-Ar radioisotopic dating. Many of the important fossils, however, were found in sediments from other widely separated localities assigned to the Ngorora Formation on the basis of either lithostratigraphic or faunal similarities (Bishop and Chapman 1970; Bishop et al. 1971; King and Chapman 1972; Bishop and Pickford 1975; Pickford 1975, 1978; Chapman et al. 1978). Here we report new lithostratigraphic, $^{40}Ar/^{39}Ar$ radioisotopic, and magnetostratigraphic

[1] Manuscript received November 6, 1989; accepted March 15, 1990.
[2] Scripps Institute of Oceanography, La Jolla, California 92093
[3] Department of Geophysical Sciences, University of Chicago, Chicago, Illinois 60637

[JOURNAL OF GEOLOGY, 1990, vol. 98, p. 567–587]
© 1990 by The University of Chicago. All rights reserved.
0022-1376/90/9804-0012$1.00

This content downloaded from 132.174.250.101 on March 28, 2019 15:13:49 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

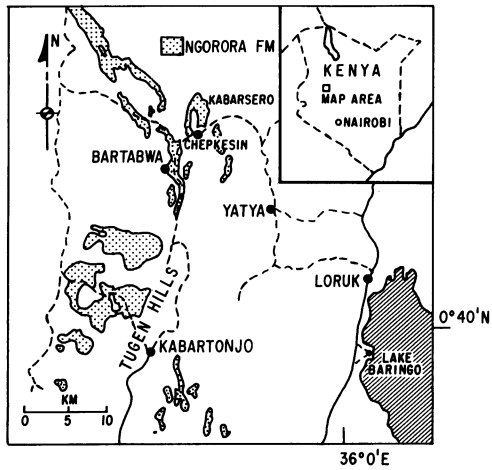


FIG. 1.—Location map showing outcrops of the Ngorora Formation (stippled areas). Dashed and solid lines are dirt and paved roads, respectively.

data for two of these outlying localities exposed near the village of Bartabwa (Hill et al. 1985). These are the Bartabwa Main Section (BB) and a shorter section (HM) in which a hominoid lower premolar was found (Hill et al. 1985). Lithostratigraphic sections were measured, and paleomagnetic and radioisotopic dating samples were collected from these areas during 1984 and 1985 field seasons.

We have also obtained new $^{40}Ar/^{39}Ar$ radioisotopic data from the type section of the Ngorora at Kabarsero, primarily from samples reported by Tauxe et al. (1985). They presented a GRTS calibration based on magnetostratigraphy and K-Ar ages from volcaniclastic strata of the type section. This calibration was approximately 10% younger than contemporaneous magnetostratigraphic sections calibrated by K-Ar ages on basalts from Iceland (McDougall et al. 1984). The new $^{40}Ar/^{39}Ar$ ages from the Ngorora Formation presented herein were obtained in part to help resolve this discrepancy.

The Neogene sedimentary and volcanic rocks discussed here are exposed along the flanks of the Tugen Hills bordering the Baringo Basin of the central Kenya Rift (fig. 1). They are dominated by thick successions of lava flows intercalated with variable thicknesses of lacustrine and fluvial sediments. The Ngorora Formation, the predominant mid-Miocene sedimentary sequence, is defined as those sedimentary and volcaniclastic rocks bounded below by the Tiim Phonolites Formation and above by the Ewalel Phonolite (Bishop and Chapman 1970).

The Tiim Phonolites Formation (~13–16 Ma; see Pickford 1978; Hill 1989; and herein) form part of a sequence of flood phonolitic lava flows that cover a large region of central and southern Kenya, centered along the Kenya rift about 130 km southeast of Bartabwa near Nakuru. These flows completely filled and overflowed the Kenya Rift and are believed to have formed an extensive tableland (Lippard 1973). At Bartabwa the contact of the base of the Ngorora Formation with Tiim Phonolites is an erosional disconformity, while at Kabarsero this contact appears to be conformable.

The Ngorora Formation consists predominantly of reworked volcaniclastic, lithoclastic, with smaller amounts of biogenic sedimentary strata. In both the Bartabwa and Kabarsero areas, the sediments fine upward, with alluvial fan deposits predominating in the bottom halves and lacustrine and nearshore lacustrine deposits forming most of the top halves of the sections. The formation thins to the west from about 450 m at Kabarsero to 200 m at Bartabwa. At Bartabwa and Kabarsero the Ngorora Formation is capped by the Ewalel flood phonolites, dated by conventional K-Ar methods at between 6.9–9.0 Ma (Chapman and Brook 1978; and herein).

RADIOISOTOPIC DATING

Methodology.—Conventional K-Ar dating of samples of Ewalel Phonolite employed separation techniques and isotope dilution procedures similar to those described by Dalrymple and Lanphere (1969) and Deino and Potts (1990).

Strata sampled for $^{40}Ar/^{39}Ar$ age determinations include pumiceous and feldspar-rich sandstones and pebbly sandstone beds, pumice tuffs, and pumice lapillistones of the Ngorora Formation, and porphyritic and aphyric lava flows of the Tiim Phonolites (table 1). The $^{40}Ar/^{39}Ar$ ages were obtained from individual K-feldspar (anorthoclase) phenocrysts, extracted from pumice whenever possible. In addition, a laser step-heat-

This content downloaded from 132.174.250.101 on March 28, 2019 15:13:49 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

TABLE 1
SUMMARY $^{40}AR/^{39}AR$ RESULTS

Sample Name	Rock Type	Strat. Ht.(m)	Polarity[a]	Age (Ma)[b]	SD Age	SEM Age	% SEM	Ave. Ca/K	±1σ Ca/K
Kabarsero									
RD-50	Tiim Phon. (wr)	0	U	13.15	0.13	0.05	0.37		
RD-21	Agglomerate	2.5	R	13.06	0.03	0.01	0.08	0.051	0.033
RD-54	Fsp. sandstone	129.0	U	12.56	0.04	0.01	0.08	0.127	0.055
RD-51	Fsp. sandstone	147.5	R	12.49	0.06	0.02	0.15	0.071	0.039
RD-53	Crystal-lith. tuff	174.0	N	12.26	0.07	0.02	0.13	0.102	0.051
RD-55	Lapilli tuff	193.8	N	12.01	0.03	0.01	0.07	0.065	0.035
RD-52	Fsp. sandstone	240.5	R	11.84[c]	0.07	0.03	0.23	0.045	0.018
RD-22	Crystal tuff	261.5	R	11.54[c]				0.150	
RD-23	Crystal tuff	290.0[d]	R	10.51	0.03	0.01	0.13	0.035	0.035
Bartabwa - BB									
TB+50	Tiim Phon. (fsp)	0	U	13.19	0.04	0.01	0.10	0.053	0.006
28/1/84A	Silty tuff	35.9	R	12.78[c]				0.089	
30/1/84H	Lapillistone	88.7	N	12.25	0.07	0.02	0.18	0.118	0.041
31/1/84A	Lapilli sandstone	126.3	U	11.78[c]	0.06	0.03	0.26	0.037	0.010
MM85-1	Lapillistone	139.6	U	11.62	0.06	0.02	0.15	0.049	0.018
1/2/84B	Fsp. sandstone	176.6	R	11.83[c]				0.023	
Bartabwa - HM									
ALD85-25	Fine lapilli tuff	1.8	N	12.97	0.06	0.02	0.12	0.057	0.055
ALD85-26	Lahar	28.2	R	12.79	0.05	0.01	0.10	0.020	0.020
ALD85-36	Lapillus	60.8	R	12.42	0.03	0.01	0.08	0.076	0.035
ALD85-35	Lapillistone	67.6	R	12.41	0.07	0.02	0.20	0.059	0.035
ALD85-29	Fsp. sandstone	68.4	R	12.30	0.06	0.01	0.12	0.055	0.037
ALD85-27	Lapilli tuff	85.7	U	12.04	0.05	0.02	0.13	0.049	0.022

[a] N = Normal, R = Reversed, U = Undetermined paleomagnetic polarity.
[b] Weighted mean ages of inferred primary populations of single-crystal feldspar determinations, except for RD-50 which is a step-heating plateau age, and except where noted by ([c]).
[c] No unequivocal primary age population; ages represent the youngest components.
[d] Position known only to within ±10-15m.
SD = 1σ weighted standard deviation.
SEM = 1σ weighted standard error of the mean (Samson and Alexander 1987)

ing experiment on a whole-rock sample of the uppermost Tiim Phonolites flow (RD-50) was performed.

Samples were prepared for single-crystal dating by gentle crushing and sieving to allow hand-selection of ~0.3–1.0 mm feldspar grains under the binocular microscope. Grains that appeared altered or cloudy; that contained large mineral, glass, or fluid inclusions; or had glass adhering to them were discarded. The purified separate was washed sequentially with 10% HCl, 5% HF, and distilled water for 5 min in an ultrasonic bath.

Samples were loaded into an aluminum holder for irradiation. The holder consisted of a stack of 16 mm diameter by 2.5 mm thick, 99.999% pure aluminum disks in which ~2 mm deep wells were drilled to accept <100 mg samples. Each disk contained 7–11 unknowns and at least one monitor standard. All samples for this study were irradiated in a single batch for 10 hr at 8 MW in the hydraulic rabbit of the Omega West research reactor of the Los Alamos National Laboratory. This reactor has a fast neutron flux of 5.7×10^{13} n/cm^2-sec. Cadmium shielding 0.032″ thick was used to reduce ^{40}Ar production through neutron interactions with K by a factor of ~20, from an unshielded $(^{40}Ar/^{39}Ar)_K$ of 0.016 ± 0.001 to a shielded value of 0.0008 ± 0.0007. During irradiation the sample package was rotated about a vertical axis

This content downloaded from 132.174.250.101 on March 28, 2019 15:13:49 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

and flipped 180° about a horizontal axis every 2 hr to improve the spatial uniformity of the neutron flux imparted to the samples.

After irradiation, the samples were transferred to a copper sample holder and loaded onto the extraction system for overnight bakeout at 200°C. The $^{40}Ar/^{39}Ar$ analyses were obtained from the fusion of individual feldspar crystals, or in the case of RD-50 from the step-heating of approximately 10 grains of whole rock phonolite. Argon was extracted by fusing the feldspars with a 6 Watt continuous Ar-ion laser beam focused to a <100 μm spot for 10–40 sec. During the step-heating experiment, the beam was widened to ~2 mm and applied for about 2 min; this process was repeated at progressively increasing power levels until the sample fused and no more gas was released. The gasses released as a result of laser heating were then scrubbed of reactive species such as CO_2, CO, and N_2 by exposure to a 150°C Zr-Fe-V alloy SAES getter for about 3 min. Argon was then measured in a Mass Analyzer Products Series 215 noble-gas mass spectrometer, operated in the static mode, using automated data-collection procedures. A typical run lasted about 20 min. Argon isotope beam intensities were measured on a Johnston electron multiplier operating at a gain of between 20,000–50,000. Typical volumes of ^{40}Ar ranged from (~0.5–10) × 10^{-10} cm^3. Blank volumes of ^{40}Ar, ^{39}Ar, ^{38}Ar, ^{37}Ar, and ^{36}Ar were (~1, 0.05, 0.02, 0.07, and 0.02) × 10^{-12} cm^3, respectively. Periodic isotopic analysis of laboratory air from an on-line stainless-steel air-pipette yielded a mean value for instrumental mass discrimination of 1.0035 ± 0.0005 per atomic mass unit [where discrimination is defined as $D = (295.5/a)^{1/4}$, in which a is the measured $^{40}Ar/^{36}Ar$ atmospheric ratio (291.4 ± 0.1)].

Sanidine from the Fish Canyon Tuff served as the monitor mineral for the $^{40}Ar/^{39}Ar$ determinations. Its reference age, 27.84 Ma, is an adjustment of the age originally reported for this sanidine by Cebula et al. (1986) in which they used the widely used hornblende reference standard MMhb-1 as their monitor. Samson and Alexander (1987) have recently revised the age of MMhb-1 to 520.4 ± 1.7 Ma. The 0.3% uncertainty in the MMhb-1 age reported Samson and Alexander (1987) is the standard error of the weighted mean age resulting from determinations of Ar and K by many laboratories. Our own preliminary calibrations of the Fish Canyon sanidine against MMhb-1 by the $^{40}Ar/^{39}Ar$ method are in agreement with the published reference age by better than 0.5%.

Between 13 and 20 single-grain $^{40}Ar/^{39}Ar$ analyses were made of each sample of the monitor standard in the irradiation package; typically these samples yielded standard errors of the mean of better than 0.1%. These analyses revealed that the neutron flux varied by <0.1% to 0.6% between levels. Consequently, the maximum age difference between grains of an otherwise homogeneous sample *within* a single level attributable to the vertical irradiation flux gradient is only ~0.07 m.y.

To account for any horizontal gradient of the neutron flux within the sample package, four aliquots of the monitor were placed at intervals along the outer margins of one of the disks. Analysis of these samples revealed a consistent flux gradient across the disk amounting to a maximum of 0.5% between extreme sample locations. Since the radial orientation of this calibration disk relative to all other disks in the irradiation stack was known, we calculated an appropriate correction to J for each of the unknowns. This correction assumes that the horizontal flux gradient throughout the irradiation package was uniform.

Errors estimates for individual laser analyses reflect errors in J and in determination of the argon isotopic ratios (Dalrymple et al. 1981). The uncertainties in the isotopic ratios were estimated by propagation of errors in D and measurement errors in Ar beam intensities for both the sample gas and blanks. Weighted mean ages for each sample were computed using the inverse variance as the weighting factor (Mandel 1964; Taylor 1982), while the standard error of the mean was calculated using the formulation of Samson and Alexander (1987). Their approach uses deviations about the weighted mean to compute a weighted uncertainty. Analytical variations from run to run and variations due to sample inhomogeneity are thus included in each error estimate. In the case of RD-50, where an "inverse" isochron correlation diagram was employed to determine an age from the step-heating experiment, the uncertainty in the

This content downloaded from 132.174.250.101 on March 28, 2019 15:13:49 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

age was obtained by combining the standard error of the intercept of the York regression (York 1969), with the standard error in J. All errors are reported at the 1 σ level of confidence.

Dating Results.—A total of 367 single-crystal feldspar analyses were obtained from 20 samples collected from the three measured stratigraphic sections (table 2). Radiogenic argon ($^{40}Ar^*$) constituted >85% of the argon released during fusion of all but eight grains. These less radiogenic feldspars were excluded from subsequent analysis because of the possibility of alteration. Individual analyses typically yielded age uncertainties between 0.03 and 0.06 m.y. (0.2 to 0.4%).

The $^{40}Ar/^{39}Ar$ analyses are plotted as age histograms in figure 2. Six samples exhibit unimodal, near-symmetrical age distributions without outliers, nine display prominent primary peaks with tails of older ages, while five show either poorly populated associations of young ages (RD-52, 31/1/84A) or scattered distributions without dominant components (RD-22, 1/2/84B, and 28/1/84A). Unimodal, near-symmetric distributions of feldspar ages within a volcaniclastic deposit suggest origination from an eruptive event that rapidly saturated the depositional environment with juvenile pyroclastic material. Older, probably detrital feldspars are often present, however, and occasionally rival the youngest age component in abundance. In such cases (e.g., RD-52 and 31/1/84A), it is uncertain whether the youngest components represent a depositional age. Nevertheless, in all but one case (1/2/84B), the youngest age component yields a mean age consistent with the observed stratigraphic sequence (table 1; fig. 2).

A population of primary volcanic feldspars in an age histogram usually is identifiable by the narrow range of ages it spans and by the symmetrical distribution of these ages about a modal value. When older detrital feldspars are also present, they usually are separated from the primary feldspars by a discernable gap in the age histogram (fig. 2). Four of the samples (RD-23, MM85-1, ALD85-27, and ALD85-25), however, do not exhibit a clear older bound for the primary age distribution. In these cases, inclusion of the grain or two that constitute the ambiguity in the primary population increases the weighted mean age by less than 0.1%, or less than 0.01 m.y.

In some cases the "primary" age population is multimodal and may represent not one but several penecontemporaneous eruptive events. In order to help resolve whether more than one event is represented in these multimodal age spectra, we compare their characteristics with those of the unimodal age spectra. The six Ngorora samples with unimodal, quasi-symmetrical age distributions (e.g., TB + 50, ALD 85-36) yield primary age populations with standard deviations of 0.02 to 0.06 m.y. (0.2 to 0.5%). Several of the multimodal spectra yield continuous or near-continuous age distributions with significantly higher standard deviations and are thus interpreted as representing two or more volcanic events. For example, ALD85-27, with a bimodal age distribution that ranges over 0.4 m.y. apparently is composed of two feldspar populations erupted 0.2 to 0.3 m.y. apart.

Ca/K ratios of individual feldspars, which are directly proportional to the $^{37}Ar/^{39}Ar$ ratios obtained during the $^{40}Ar/^{39}Ar$ age determinations, can occasionally be used to help differentiate multiple volcanic events represented within a single sample. For example, analyses of RD-23 yielded one feldspar (1270-03 in table 2) somewhat older than the oldest easily recognized primary feldspar, but with the unusually low Ca/K ratio (0.012) characteristic of the primary population (0.035 ± 0.035). Other older feldspars in this sample had distinctly higher Ca/K ratios (~0.11). On the basis of this comparison, feldspar 1270-03 was included in the age calculation for this sample while the other older feldspars were excluded. Unfortunately, the ability of Ca/K to characterize contaminants is limited in the Ngorora suite, because these ratios vary by about an order of magnitude within a single eruptive event, while different eruptive events often yield similar average Ca/K ratios. Weighted mean ages, average Ca/K ratios, and additional analytical statistics for the inferred primary age components of each sample are listed in table 1.

The mean age, Ca/K ratio, and position within the local paleomagnetic stratigraphy of the dated stratigraphic units can be used to correlate equivalent volcaniclastic horizons in different localities. One pair of samples (30/1/84H and RD-53) is correlated on the strength of similar, atypically high Ca/K

This content downloaded from 132.174.250.101 on March 28, 2019 15:13:49 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

TABLE 2

EXAMPLE $^{40}AR/^{39}AR$ ANALYTICAL DATA, NGORORA FORMATION, BARINGO, KENYA.

Lab ID#	Ca/K	$^{36}Ar/^{39}Ar$	$^{40}Ar^*/^{39}Ar$	^{40}Ar (Moles × 10^{14})	$\%^{40}Ar^*$	Age (Ma) ±1σ[a]	
RD-53 (J=0.007262 ± 0.000005)							
1299-14	0.112	0.00005	0.930	0.57	98.6	12.15	0.15
1299-02	0.101	0.00003	0.932	2.1	99.3	12.17	0.03
1299-12	0.068	0.00003	0.934	1.4	99.2	12.19	0.03
1299-18	0.135	0.00003	0.934	1.3	99.3	12.20	0.03
1299-19	0.092	0.00012	0.935	2.0	96.5	12.20	0.03
1299-17	0.046	0.00003	0.935	1.0	99.1	12.21	0.03
1299-01	0.085	0.00007	0.936	1.8	98.1	12.22	0.03
1299-13	0.114	0.00003	0.936	1.4	99.3	12.22	0.03
1299-11	0.108	0.00003	0.937	0.91	99.3	12.24	0.04
1299-09	0.124	0.00008	0.937	1.4	97.9	12.24	0.03
1299-15	0.222	0.00005	0.940	0.82	99.3	12.27	0.03
1299-06	0.063	0.00002	0.940	1.6	99.3	12.27	0.03
1299-16	0.133	0.00050	0.940	2.1	86.8	12.28	0.04
1299-03	0.051	0.00018	0.941	1.2	94.7	12.29	0.04
1299-10	0.206	0.00004	0.943	1.2	99.4	12.32	0.03
1299-07	0.032	0.00002	0.945	1.3	99.2	12.34	0.03
1299-08	0.089	0.00012	0.948	1.8	96.7	12.38	0.03
1299-04	0.042	0.00003	0.949	1.0	99.1	12.39	0.03
					Wtd. Ave.=	12.26	0.02
Anomalously old:							
1299-05	0.102	0.00002	0.968	1.7	99.5	12.63	0.04
1299-20	0.038	0.00011	0.973	1.0	96.6	12.71	0.17
RD-23 (J=0.007280 ± 0.000005)							
1270-04	0.046	0.00009	0.7992	1.2	96.8	10.47	0.02
1270-01	0.113	0.00045	0.7993	1.2	85.9	10.47	0.03
1270-14	0.022	0.00035	0.7993	1.7	88.5	10.47	0.03
1270-05	0.020	0.00003	0.8002	1.2	98.7	10.48	0.02
1270-02	0.020	0.00037	0.8017	2.2	87.8	10.50	0.03
1270-13	0.079	0.00028	0.8018	1.5	90.8	10.50	0.03
1270-07	0.019	0.00006	0.8027	1.6	97.7	10.51	0.02
1270-06	0.013	0.00002	0.8033	1.8	99.1	10.52	0.02
1270-10	0.013	0.00002	0.8044	1.7	99.1	10.53	0.02
1270-03	0.012	0.00022	0.8102	1.8	92.4	10.61	0.03
					Wtd. Ave.=	10.51	0.01
Anomalously old:							
1270-15	0.121	0.00003	0.8269	1.9	99.4	10.83	0.03
1270-09	0.101	0.00004	0.8311	1.3	98.9	10.88	0.03
1270-08	0.046	0.00004	1.0296	1.3	99.0	13.47	0.03
MM85-1 (J=0.007242 ± 0.00001)							
1284-07	0.081	0.00014	0.8844	0.52	95.5	11.52	0.04
1284-11	0.064	0.00005	0.8867	0.25	98.5	11.55	0.06
1284-10	0.022	0.00010	0.8874	0.51	96.7	11.56	0.04
1284-05	0.050	0.00008	0.8877	0.35	97.4	11.56	0.06
1284-14	0.057	0.00014	0.8879	0.29	95.6	11.56	0.06
1284-03	0.065	0.00040	0.8917	0.35	88.4	11.61	0.09
1284-01	0.033	0.00012	0.8919	0.79	96.2	11.62	0.04
1284-06	0.044	0.00004	0.8926	0.52	98.7	11.63	0.04
1284-08	0.061	0.00006	0.8930	0.96	98.2	11.63	0.03
1284-04	0.027	0.00009	0.8955	0.49	96.9	11.66	0.05
1284-09	0.046	0.00012	0.8960	0.52	96.2	11.67	0.04
1284-13	0.023	0.00006	0.8964	0.13	98.1	11.68	0.12
1284-12	0.040	0.00018	0.8972	0.55	94.2	11.69	0.04
1284-02	0.061	0.00003	0.9035	0.19	99.1	11.77	0.09
1284-15	0.065	0.00007	0.9076	0.27	98.0	11.82	0.07
					Wtd. Ave.=	11.62	0.02

[a] Errors in age quoted for individual runs are 1σ analytical uncertainty; errors in age quoted for weighted means are 1σ standard error of the mean. Ca/K is calculated from $^{37}Ar/^{39}Ar$ using a divisor of 0.51. $^{40}Ar^*$ refers to radiogenic argon. 'Moles ^{40}Ar' refers to the estimated total moles of ^{40}Ar released during fusion based on spectrometer sensitivity considerations. $\lambda = 5.543 \times 10^{-10}$ y^{-1}. Isotopic interference corrections: $(^{36}Ar/^{37}Ar)_{Ca} = 2.58 \times 10^{-4} \pm 6 \times 10^{-6}$, $(^{39}Ar/^{37}Ar)_{Ca} = 6.7 \times 10^{-4} \pm 3 \times 10^{-5}$, $(^{40}Ar/^{39}Ar)_K = 8 \times 10^{-4} \pm 7 \times 10^{-4}$. Analyses for each sample are arranged in order of increasing age.

This content downloaded from 132.174.250.101 on March 28, 2019 15:13:49 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

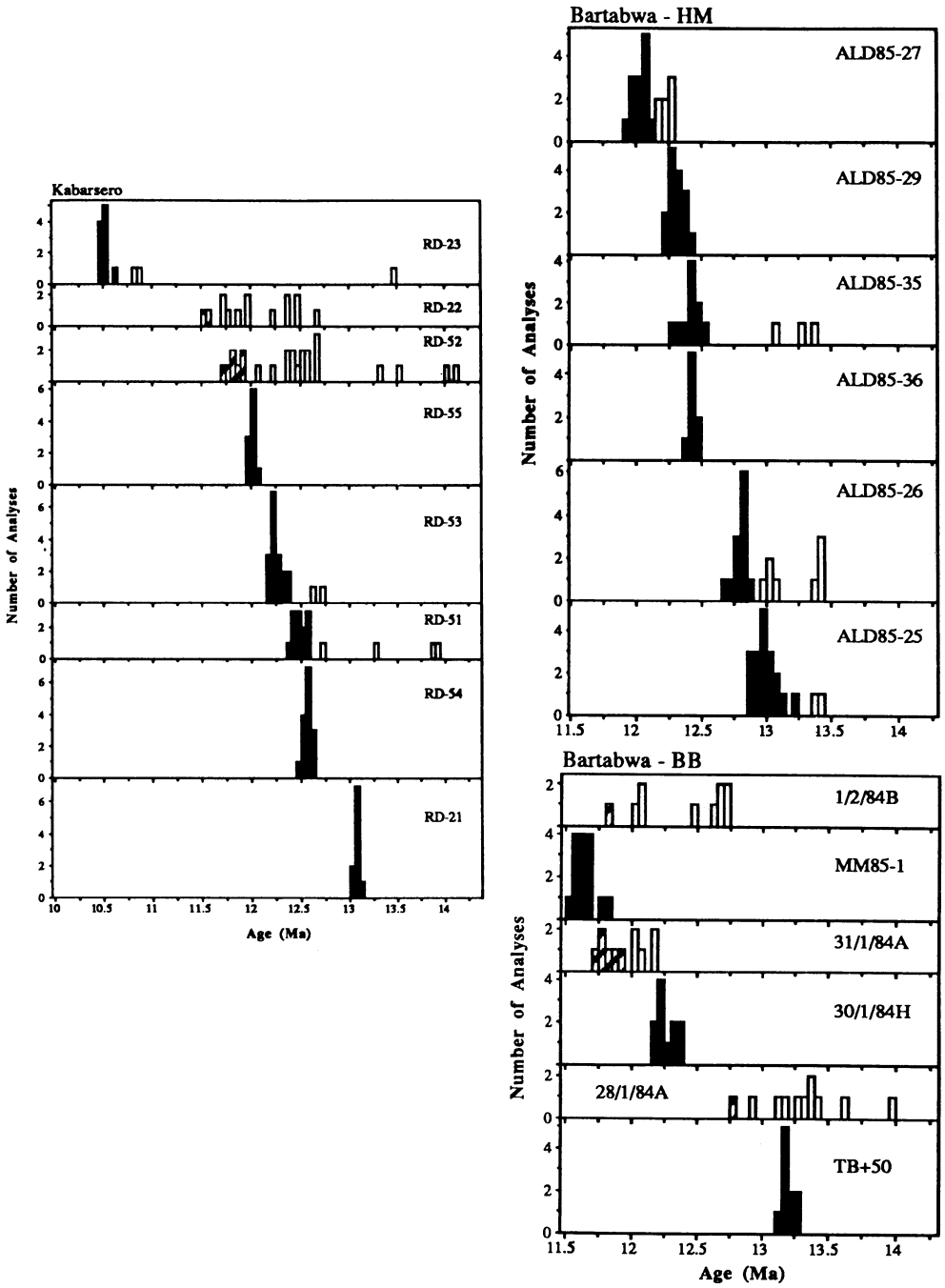


FIG. 2.—Histograms of single-crystal $^{40}Ar/^{39}Ar$ age determinations with $\%^{40}Ar^*$ greater than 85%. Black-filled analyses represent inferred primary volcanic components; stripped analyses are questionable primary feldspars; clear analyses are inferred contaminants. Bins are 0.05 m.y. wide. Samples are arranged in stratigraphic order. (*a*) Kabarsero. (*b*) Bartabwa main section (BB). (*c*) Bartabwa hominoid section (HM).

This content downloaded from 132.174.250.101 on March 28, 2019 15:13:49 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

TABLE 3

$^{40}AR/^{39}AR$ ANALYTICAL DATA FOR STEP-HEATING OF WHOLE-ROCK TIIM PHONOLITE

Lab ID#	Laser Pwr. (Watts)	Ca/K	$^{36}Ar/^{39}Ar$	$^{40}Ar^*/^{39}Ar$	^{40}Ar (Moles × 10^{14})	$\%^{40}Ar^*$	Age (Ma) ±1σ	
RD-50 (J=0.007281 ± 0.000005)								
1301-01A	0.1	0.249	0.19691	0.081	0.6	0.1	1.06	82.12
1301-01B	0.3	0.157	0.02676	0.614	15.0	7.2	8.04	1.00
1301-01C	0.4	0.084	0.01098	0.939	14.0	22.4	12.28	0.56
1301-01D	0.6	0.054	0.00344	1.035	22.0	50.5	13.54	0.18
1301-01E	1.0	0.041	0.00266	1.026	4.2	56.6	13.43	0.15
1301-01F	1.5	0.139	0.00142	1.022	6.3	71.0	13.37	0.10
1301-01G	2.0	0.177	0.00143	1.015	8.6	70.9	13.28	0.07
1301-01H	2.5	0.148	0.00158	0.997	5.6	68.3	13.04	0.10
1301-01I	3.0	0.121	0.00164	1.012	3.1	67.8	13.24	0.20
1301-01J	3.5	0.206	0.00180	0.992	5.2	65.4	12.98	0.17
1301-01K	4.0	0.197	0.00190	1.025	3.7	64.9	13.41	0.15
1301-01L	4.5	0.228	0.00188	1.028	2.9	65.2	13.46	0.18
1301-01M	5.0	0.215	0.00211	1.033	2.0	62.6	13.52	0.22
1301-01N	5.5	0.185	0.00220	1.013	1.5	61.1	13.26	0.28
1301-01O	6.0	0.212	0.00216	1.016	2.7	61.6	13.30	0.17
					Plateau =		13.34	0.09
					Total Gas =		13.06	0.23

NOTE.—Sample is from the Tiim Phonolite at the base of the Ngorora Formation, Kabarsero section. It is an aphyric phonolite with a microcrystalline groundmass of alkali feldspar and sodic pyroxene with ~25% intersertal devitrified glass. Other notations same as Table 2.

ratios, virtually identical ages, and similar position within magnetozones of normal polarity. This horizon provides an important stratigraphic link between the Bartabwa and Kabarsero sections.

The results of a laser step-heating experiment on a whole-rock sample from the uppermost Tiim Phonolite underlying the Ngorora Formation at Kabarsero (RD-50; table 3) suggest that Ngorora sedimentation began shortly after phonolite emplacement. The $^{40}Ar/^{39}Ar$ age spectrum for this experiment (fig. 3) shows anomalously young ages obtained for the initial three steps, indicating partial disturbance of the argon systematics, probably as a result of low-temperature alteration. Nevertheless, the remaining steps delineate a flat portion of the spectrum, of which a four-step segment (D–G) meets the criteria of Fleck et al. (1977) and Lanphere and Dalrymple (1978) for a valid plateau. This plateau yields a weighted mean age of 13.34 ± 0.09 Ma, compared to a integrated (total fusion) age of 13.06 ± 0.23 Ma for this experiment. An isochron formed from the four plateau steps is concordant with the plateau age, giving 13.15 ± 0.13 Ma, with MSWD = 0.378 (an index of the goodness of fit, which indicates in this case that all dispersion may be accounted for by analytical uncertainties [McIntyre et al. 1966; McDougall and Harrison 1988]). The $^{40}Ar/^{36}Ar$ intercept of the isochron is 304 ± 5, within 2σ of the atmospheric ratio of 295.5. We believe the isochron provides a better estimate of the age of this flow because it makes no assumptions concerning the composition of the trapped

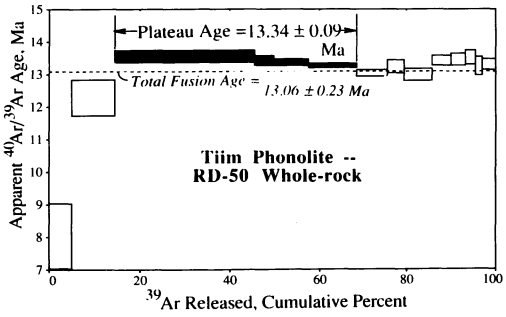


FIG. 3.—$^{40}Ar/^{39}Ar$ age-spectrum for laser step-heating of whole-rock Tiim Phonolite from the Kabarsero section.

This content downloaded from 132.174.250.101 on March 28, 2019 15:13:49 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

TABLE 4
NEW CONVENTIONAL K-AR DATING RESULTS FOR THE EWALEL PHONOLITES AND NGORORA FORMATION, KENYA.

Berkeley No.	ANU No.	Material Dated	K (Wt.%)	Weight (g)	$^{40}Ar^*$ (10^{-11} mol/g)	$\%^{40}Ar^*$	Age (Ma ± 1σ)
Ewalel Phonolite:							
1/2/84C[a] (KA 4954)	--	Groundmass	4.700, 4.758, 4.730	3.312	5.965	68.4	7.26 ± 0.17
22/1/84A[b] (KA 4845)	--	Groundmass	5.025, 4.882, 4.928	4.839	6.532	60.5	7.60 ± 0.14
Ngorora Formation:[c]							
RD-21	87-40	Anorthoclase	5.421, 5.461	0.205	12.71	89.9	13.42 ± 0.14
				0.196	12.63	88.2	13.34 ± 0.16
					Mean Age=		13.38 ± 0.15
RD-51	87-41	Anorthoclase	5.507, 5.538	0.173	12.10	86.2	12.60 ± 0.13
				0.194	12.06	82.1	12.54 ± 0.13
					Mean Age=		12.57 ± 0.13
RD-52	87-42	Anorthoclase	5.633, 5.621	0.210	11.62	87.0	11.87 ± 0.14
				0.195	11.79	86.0	12.04 ± 0.13
					Mean Age=		11.96 ± 0.14

NOTE.—$\lambda_\varepsilon + \lambda_{\varepsilon'} = 0.581 \times 10^{-10}/a$; $\lambda_\beta = 4.962 \times 10^{-10}/a$; $^{40}K/K = 1.167 \times 10^{-4}$ mol/mol.
[a]Sparsely porphyritic phonolite with a few percent phenocrysts of anorthoclase in a trachytic groundmass with intersertal microcrystalline alkali feldspar, patchy sodic pyroxene, opaque minerals, and ~30% glass.
[b]Similar in petrographic characteristics to 1/2/84C except with ~20% glass.
[c]Data on the Ngorora Formation provided by I. McDougall at Australian National University.

non-radiogenic component, whereas the plateau-age calculation assumes that the trapped component has the same composition as the modern atmosphere.

An upper age limit to the period of Ngorora sedimentation is provided by two new conventional K-Ar determinations on groundmass separates of the Ewalel Phonolite overlying the Ngorora Formation at Kabarsero and at Bartabwa (BB). These yielded ages of 7.60 ± 0.14 Ma at Kabarsero (22/1/84A, table 4) and 7.26 ± 0.17 Ma at Bartabwa (1/2/84C). These analyses are more consistent in age, and have higher $^{40}Ar^*$ content (60–68%) than previously published K-Ar ages for these flows in the Baringo Basin (17–48% $^{40}Ar^*$; 6.9–9.0 Ma; Chapman and Brooks 1978). We thus prefer to use these new determinations to represent the age of the Ewalel Phonolites at the Kabarsero and Bartabwa sections.

Finally, three new K-Ar age determinations of the same anorthoclase concentrates from the Ngorora Formation discussed in Tauxe et al. (1985) were analyzed at the Australian National University (ANU) by I. McDougall. These analyses are discussed below in the Discussion and Conclusions section.

PALEOMAGNETIC RESULTS

Paleomagnetic sampling and measurement procedures used in this study of the BB and HM stratigraphic sections at Bartabwa are nearly identical to those used previously for the Kabarsero section (Tauxe et al. 1985). At least three specimens were collected from each of 145 sites using the techniques developed by Johnson et al. (1975). From more than half of the sites, one of these specimens was selected for step-wise thermal demagnetization. The demagnetization behavior of representative normal and reversed speci-

This content downloaded from 132.174.250.101 on March 28, 2019 15:13:49 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

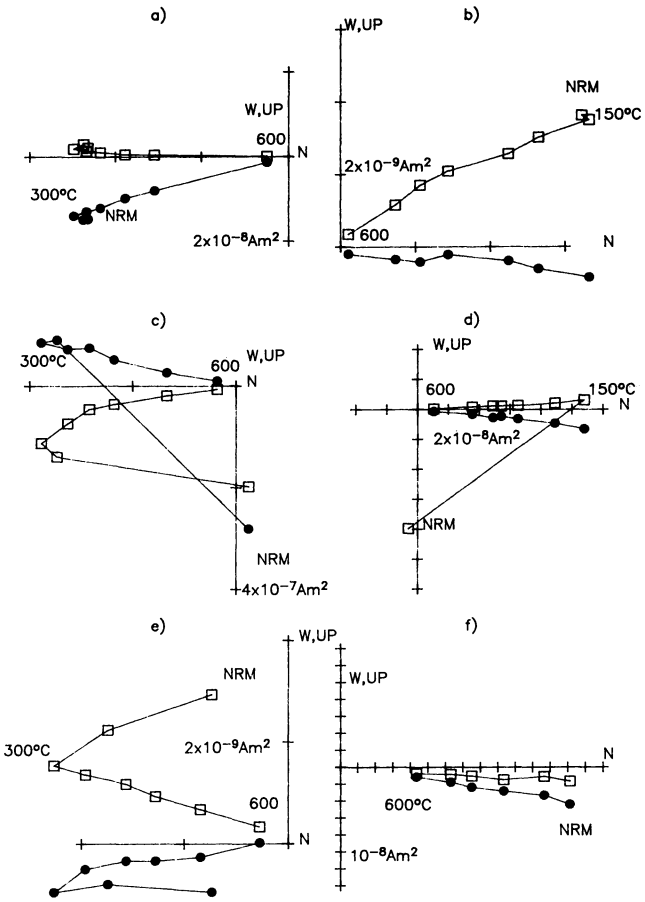


FIG. 4.—Vector end-point diagrams from representative normal (right) and reversed (left) specimens from the (*a*) and (*b*) Bartabwa hominoid section (HM), (*c*) and (*d*) Bartabwa main section (BB), and (*e*) and (*f*) the Kabarsero sections (KN, KS). Solid symbols are projections onto the horizontal plane and open symbols are projections onto the North-South vertical plane. Temperature steps were: NRM, 150, 300, 400, 450, 500, 550 and 600°C for all test specimens.

mens from each area is shown on vector end-point diagrams in figure 4. Many specimens had blocking temperatures in excess of 600°C, suggesting the presence of hematite despite the lack of red pigmentation. A characteristic direction was isolated in most specimens by ~300°C; hence the remaining specimens were treated to a single thermal demagnetization step.

Site means and confidence parameters were calculated for the three specimens using the methods of Fisher (1953). Those that exhibited random directions using the criteria of Watson (1956), and those with less than three surviving specimens were eliminated from further interpretation, leaving 106 sites or 73% of the total sampled. Statistically significant site means from the three sections are plotted on an equal-area projection in figure 5. The very slight structural dips of less than 20° preclude a significant fold test; furthermore the geographic and dip-corrected data are quite similar.

Figure 6 shows the combined data for the three sections. These have been analyzed using the statistical method of Kylstra and Tauxe (1989), which employs a data resampling technique known as the bootstrap (Efron 1982), substituting repetitive calculations for analytical solutions in order to estimate 95% confidence ellipses for the combined data sets. The procedure involves random selection of data points to generate different "pseudo samples" derived from the same

This content downloaded from 132.174.250.101 on March 28, 2019 15:13:49 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

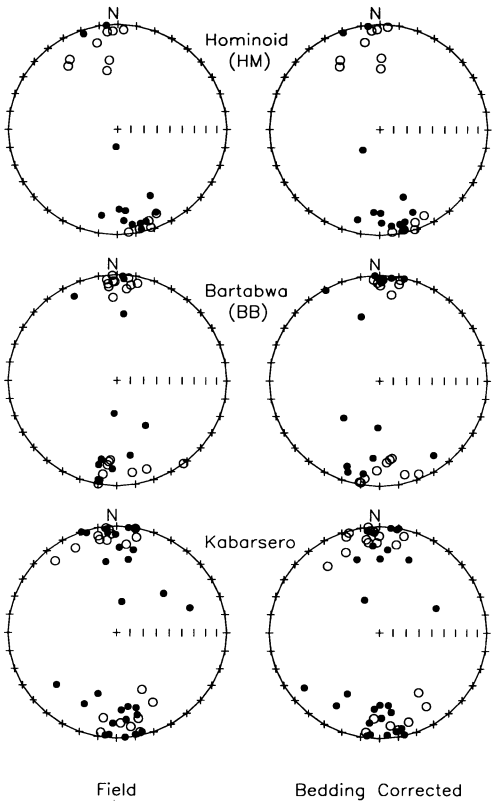


FIG. 5.—Equal-area projections of statistically significant site mean directions before (left) and after (right) correction for bedding dip. Solid symbols are lower hemisphere and open symbols are upper hemisphere projections.

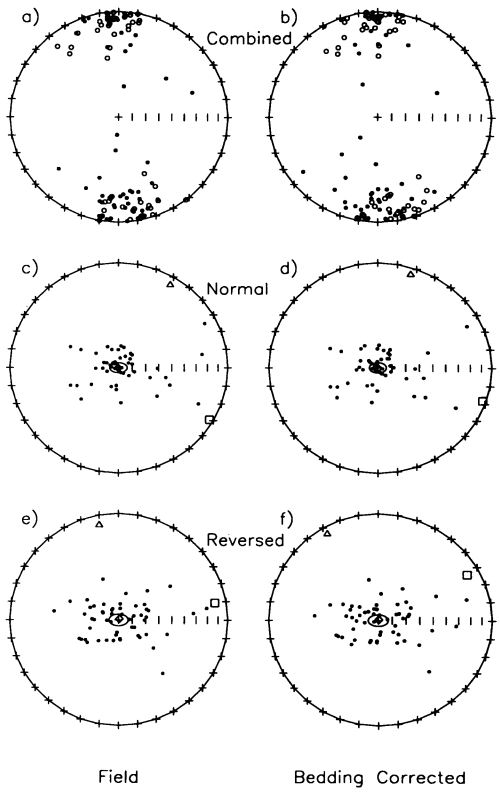


FIG. 6.—Equal-area projections of the combined data from all three sections. (*a*) Conventional projection, with directions corrected for sample orientation. (*b*) Conventional projection with dip-corrected data. (*c*) Data from 6*a* assigned to the normal mode projected along the principal direction of the data set. North (diamond), east (triangle) and down (square) are plotted for reference. The 95% confidence ellipse (see table 5) is also shown. (*d*) Same as 6*c* for data from 6*b*. (*e*) Data from 6*a* assigned to the reversed mode, otherwise same as 6*c*. (*f*) Data from 6*b* plotted same as 6*c*.

population. A mean and a confidence ellipse that includes 95% of the "bootstrapped" means is then calculated for each pseudo sample. The fundamental assumption is that the data set is sufficiently large to represent the underlying distribution, even though the form of that distribution is not known.

The statistical parameters calculated for the Ngorora data are presented in table 5. The mean directions agree well with the expected direction for these sections (located at 1°N and 35.5°E) and are nearly parallel to the present dipole field (inclined downward by about 2°). The combined data are plotted in the conventional manner (with respect to north, east, and down) and corrected for sample orientation in figure 6*a* and for dip attitudes in figure 6*b*. The data comprising the normal modes are plotted relative to the principal components of the data set in figure 6*c* (corrected for sample orientation) and 6*d* (dip corrected). The center of the diagram is the first principal component (approximately parallel to the mean vector direction of the data). The reversed group of data are plotted similarly in figure 6*e* and 6*f*. The normal and reversed groups are antipodal within the confidence regions given in table 5.

Lithostratigraphic and magnetostratigraphic data are plotted as Virtual Geomagnetic Pole (VGP) positions in figure 7. VGP latitudes for individual specimens are plotted as circles, and sites whose directions are significantly clustered (using the criteria of Watson 1956) are plotted as squares and con-

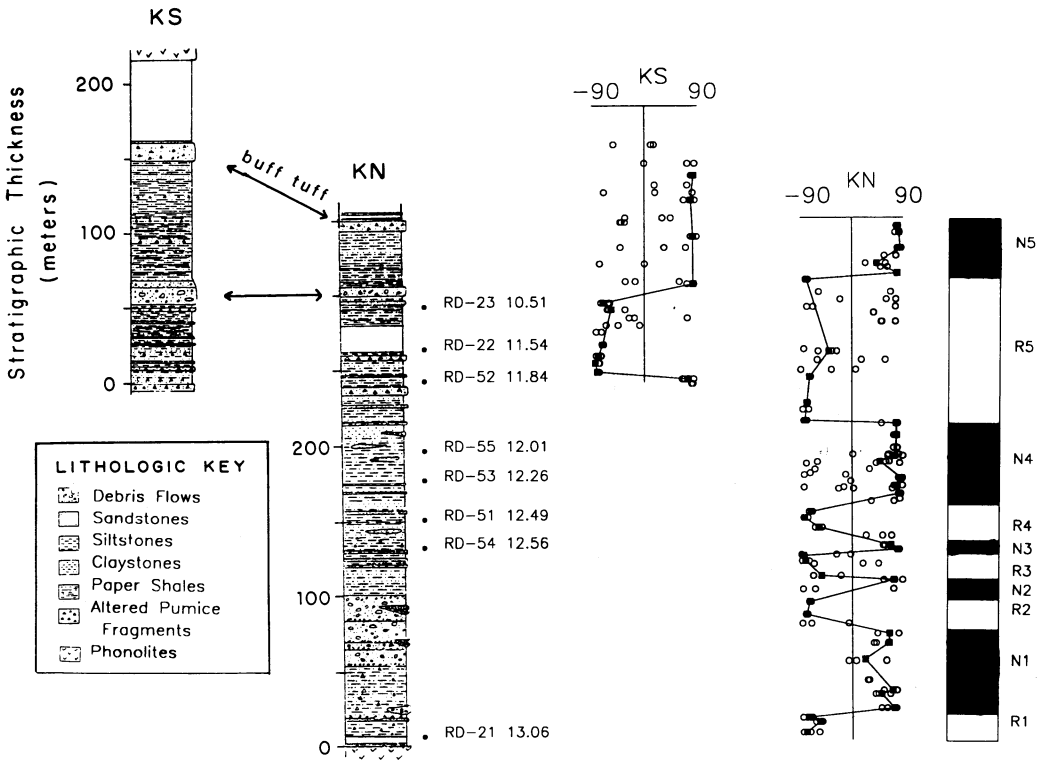
This content downloaded from 132.174.250.101 on March 28, 2019 15:13:49 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

TABLE 5

SUMMARY OF PALEOMAGNETIC STATISTICS FOR DIRECTIONAL DATA

Section	Mode	N	Mean Decl.	Mean Incl.	Semi-axes of 95% confidence ellipses ζ_{95}	ζ_{95} Decl.	ζ_{95} Incl.	η_{95}	η_{95} Decl.	η_{95} Incl.
Kabarsero	Normal	25	359.7	6.5	6.6	85	-37	12.1	9	53
	Reverse	22	177.6	5.0	6.1	85	25	11.4	278	64
Bartabwa	Normal	15	359.5	-3.1	4.9	89	45	7.6	293	81
	Reverse	16	179.1	6.2	8.2	91	-5	14.4	52	83
Hominoid	Normal	10	347.0	-15.8	8.2	49	60	12.0	85	-25
	Reverse	16	168.0	12.2	4.6	76	10	10.3	301	75
Combined (field)	Normal	50	357.1	-1.1	3.9	88	-31	6.5	81	59
	Reverse	54	175.2	7.6	4.4	84	10	7.1	295	78
(bedding)	Normal	50	357.3	-1.5	4.0	88	-19	6.1	79	72
	Reverse	54	177.2	7.5	4.3	83	29	7.2	301	59

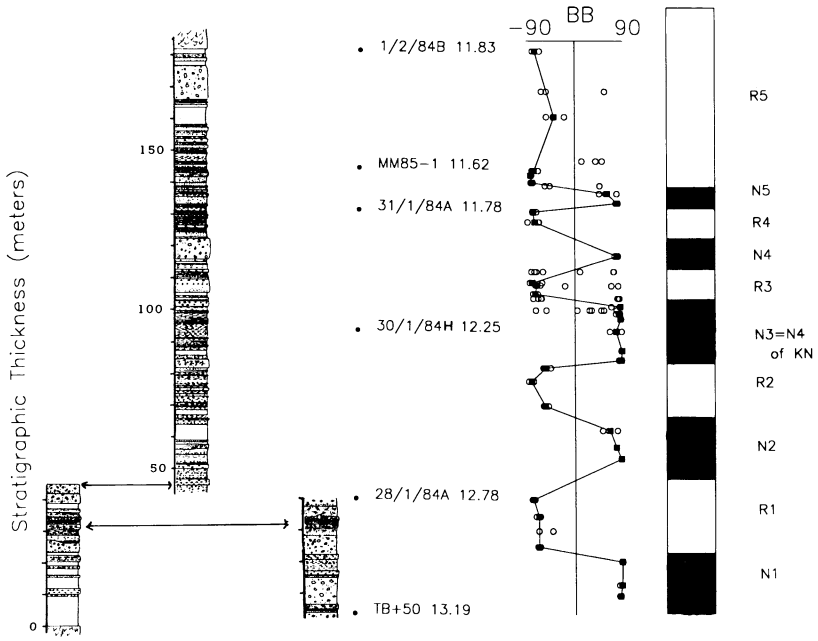
NOTE.—Directions from individual sections were corrected for sample orientation. Both field and bedding corrected data are given for the combined data set.



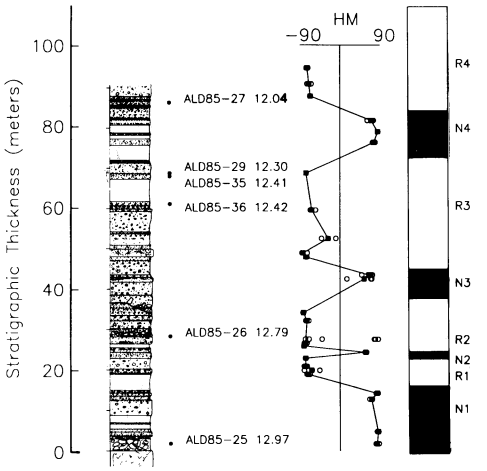
A

FIG. 7.—Lithostratigraphic sections with corresponding $^{40}Ar/^{39}Ar$ eruption ages and magnetostratigraphy. Paleomagnetic data are plotted as VGP latitudes. Circles represent individual specimen directions and squares connected by lines are statistically significant site means. Polarity interpretations are shown to the right (black = normal, white = reversed). (*a*) Kabarsero north and south (KN and KS), as in Tauxe et al. (1985). (*b*) The Bartabwa main section (BB).

This content downloaded from 132.174.250.101 on March 28, 2019 15:13:49 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).



B



C

FIG. 7.—Continued. (*c*) The Bartabwa hominoid section (HM).

nected by lines. These form the basis for the polarity interpretations shown to the right of figure 7 and allow correlation of these magnetostratigraphic data to the GRTS. The polarity interpretations for each section are labeled N1, R1, N2, R2, etc., representing normal and reversed magnetozones counted upward from the base of the section.

DISCUSSION AND CONCLUSIONS

Correlation of Magnetostratigraphic Sections.—The ambiguity inherent in correlating a local magnetostratigraphy to the GRTS usually allows for more than one possible interpretation. Factors that may constrain these alternatives include the degree to which the magnetostratigraphic pattern matches the GRTS and its consistency with other available age information. Rarely does any one correlation satisfy these constraints completely; nevertheless several usually do not require unreasonable violations. The GRTS we use as our reference is that of Berggren et al. (1985), hereinafter called BKFV85.

Table 6 summarizes the three likeliest correlations of the magnetostratigraphic patterns to the GRTS, labeled (I), (II), and (III). Based on these correlations, a GRTS "age" at the stratigraphic level of each confidently dated volcaniclastic horizon was calculated by interpolation between adjacent paleomagnetic boundaries, assuming constant sediment accumulation rates (table 6). We then calculated the difference, Δ, between the $^{40}Ar/^{39}Ar$ and GRTS ages (table 7). The lack of fit of the isotopic age data to the GRTS is partially reflected by the sum of the squares of Δ; hence the best correlation is (I), for which the

This content downloaded from 132.174.250.101 on March 28, 2019 15:13:49 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

TABLE 6
POSSIBLE CORRELATIONS OF MAGNETOSTRATIGRAPHY TO THE TIME SCALE

Section	Boundary	Stratigraphic Position (m)	(I) Age (Ma)	(II) Age (Ma)	(III) Age (Ma)	(I*) Age (Ma)
KN	N1o	18.0	13.01	13.01	13.46	13.33
	N1y	74.5	12.83	12.83	13.20	13.13
	N2o	94.8	12.62	12.62	13.01	12.89
	N2y	108.5	12.58	12.58	12.83	12.84
	N3o	125.5	12.49	12.49	?	12.74
	N3y	134.1	12.46	12.46	?	12.70
	N4o	158.5	12.12	12.12	12.12	12.33
	N4y	213.0	11.86	11.55	11.55	12.03
	N5o	310.3	10.42	10.42	10.42	10.42
BB	N1y	19.0	12.83	12.83	13.69	13.13
	N2o	42.3	12.62	12.62	13.46	12.89
	N2y	61.5	12.46	12.46	13.20	12.70
	N3o	78.4	12.12	12.12	13.01	12.33
	N3y	98.5	11.86	11.86	12.83	12.03
	N4o	108.1	?	11.73	12.62	?
	N4y	117.5	?	11.55	12.56	?
	N5o	126.9	?	?	12.49	?
	N5y	133.5	11.55	?	12.46	11.69
HM	N1y	16.6	12.83	12.83	12.83	13.13
	N2o	23.6	12.62	?	?	12.89
	N2y	25.1	12.58	?	?	12.84
	N3o	38.2	12.49	12.62	12.62	12.74
	N3y	45.5	12.46	12.58	12.58	12.70
	N4o	72.5	12.12	12.49	12.49	12.33
	N4y	84.3	11.86	12.46	12.46	12.03

NOTE.—The boundaries are the magnetozones labelled in Fig. 7. The subscripts (o) and (y) are older and younger respectively and the stratigraphic positions are the mid-point between significant sites in meters above base. The ages for the three alternative correlations discussed in the text (I, II and III) are calculated from the Berggren *et al.* (1985) time scale. The ages labelled (I*) are for a modified time scale as described in the text.

sum of the squares of the Δ values is minimized.

The preferred correlation (I) minimizes the differences between the isotopic and GRTS ages, but also requires some mismatch between the local magnetostratigraphic patterns and the GRTS (fig. 8). The correlation of the type section at Kabarsero, KN, to the GRTS made here is slightly different than that reported previously by Tauxe et al. (1985). This reflects the inclusion by Tauxe et al. (1985) of one site (in what we now label magnetozone N4) that does not meet our present acceptance criteria and thus was eliminated from the analysis. Our correlation indicates that several short sub-chrons are missing in the section, which is reasonable given the number of sites with random data. Although the first normal interval in the Kabarsero section, N1, appears overly thick, its thickness is consistent with the lithological variations discussed earlier and in Tauxe et al. (1985), and with the $^{40}Ar/^{39}Ar$ ages, which indicate that the sediment accumulation rate was high while this part of the section was being deposited. HM section at Bartabwa is a short section and may be confidently correlated as shown, even though the fourth normal interval, N4, appears too thin. The match of the BB section at Bartabwa to the GRTS using paleomagnetic criteria alone is best for correlation (III) in table 6; this correlation, however, indicates that the isotopic ages for the section are ~0.6–0.8 m.y. too old. This discrepancy is an order of magnitude larger than the typical analytical uncertainty of the $^{40}Ar/^{39}Ar$ dating, and therefore correlation (III) is considered unlikely. Furthermore, the first

This content downloaded from 132.174.250.101 on March 28, 2019 15:13:49 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

TABLE 7
SUMMARY OF AGE INFORMATION

Sample	Strat.Pos. (m)	$^{40}Ar/^{39}Ar$ Age	(I) Age	Δ	(II) Age	Δ	(III) Age	Δ	(I*) Age	Δ
KN										
RD-54	129.0	12.56	12.37	0.19	12.37	0.19	12.59	-0.03	12.72	-0.16
RD-51	147.5	12.49	12.27	0.22	12.27	0.22	12.28	0.21	12.49	0.00
RD-53	174.0	12.26	12.05	0.21	11.95	0.31	11.95	0.31	12.24	0.02
RD-55	193.8	12.01	11.95	0.06	11.75	0.26	11.75	0.26	12.14	-0.13
BB										
30/1/84H	88.7	12.25	11.99	0.26	11.99	0.26	12.92	-0.67	12.18	0.07
MM85-1	139.6	11.62					12.49	-0.87		
HM										
ALD85-26	28.2	12.79	12.56	0.23	12.72	0.07	12.72	0.07	12.82	-0.03
ALD85-36	60.8	12.42	12.30	0.12	12.54	-0.12	12.54	-0.12	12.53	-0.11
ALD85-35	67.6	12.41	12.24	0.17	12.52	-0.11	12.52	-0.11	12.46	-0.05
ALD85-29	68.4	12.30	12.23	0.07	12.52	-0.22	12.52	-0.22	12.45	-0.15
ALD85-27	85.7	12.04	11.78	0.26	12.45	-0.41	12.45	-0.41	12.01	0.03
$\Sigma(\Delta^2)$				0.37		0.56		1.66		0.09

NOTE.—$^{40}Ar/^{39}Ar$ ages are from Table 1. Magnetic age estimates are based on various correlations to the time scale: (I) preferred correlation to BKFV85 shown in Fig. 8 and Table 6. (II) and (III) are alternate correlations as listed in Table 6. (I*) are ages calculated using the preferred correlation to the GRTS (I), but using the revised ages discussed in the text. Δ is the isotopic minus the magnetic age and $\Sigma(\Delta^2)$ is the sum of the squares of these differences. Ages are given in Ma.

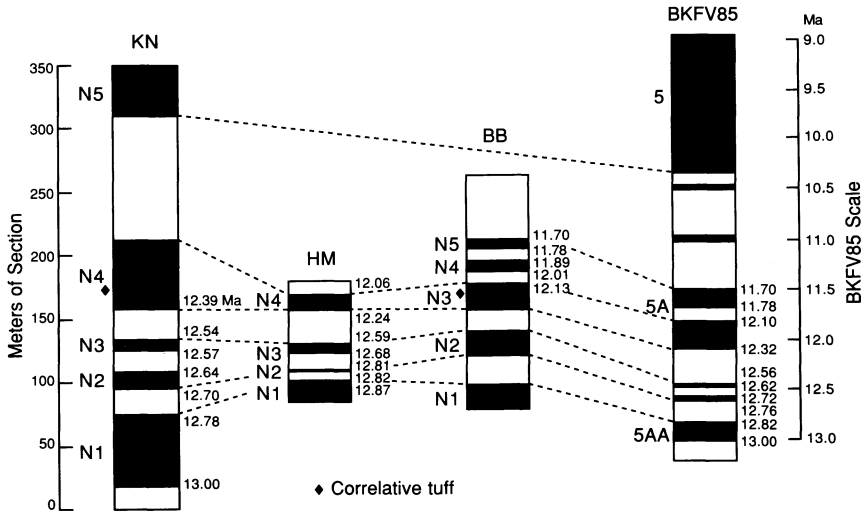


FIG. 8.—Preferred correlation of the magnetostratigraphic results to the GRTS. The polarity logs for each section are scaled as meters of section; the BKFV85 log is scaled as Ma. Interpolated $^{40}Ar/^{39}Ar$ isotopic ages of magnetozone boundaries (in Ma) are given along the right of each log. For the BKFV85 log, boundary ages are averages of the three Ngorora Formation sections, and serve as tie-points for calibration of the GRTS. Diamond identifies the position of correlative tuff RD-53 in the KN section with 30/1/84H in the BB section.

This content downloaded from 132.174.250.101 on March 28, 2019 15:13:49 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

normal interval in the BB section, N3, and the fourth normal in the KN section, N4, are linked by the probable correlation of tuff horizons 30/1/84H at BB and RD-53 at KN. The basal strata overlying the widespread plateau Tiim Phonolites are likely also roughly contemporaneous; hence the first normal intervals, N1's, in both the KN and BB sections probably represent the same magnetic polarity interval in the GRTS. These considerations effectively constrain the magnetostratigraphic correlations to those shown in figure 8. They indicate that the second normal interval in the BB section, N2, represents two short normal subchrons and one reversed subchron in the GRTS, and that the fourth interval of normal polarity in the BB section, N4 (which is defined by only one sample site), is a short subchron not recognized in the GRTS. The paleomagnetic reversal ages defined in this manner and the isotopic ages discussed earlier are plotted together against stratigraphic position for each of the three measured sections in figure 9.

GRTS Age Calibration.—The difference between the $^{40}Ar/^{39}Ar$ ages and the "GRTS ages," Δ, averages 0.18 m.y. (1.4%). All of the isotopic ages are older than their corresponding magnetochronologic ages. The systematic discrepancy could be explained by: (1) the correlation of the local magnetic stratigraphy to the GRTS being systematically displaced; (2) the $^{40}Ar/^{39}Ar$ isotopic ages not representing the time of deposition of the dated strata; (3) the calibration of the GRTS not being correct within this time interval, or (4) differences in standardization procedures used by various K-Ar laboratories. The first explanation produces a far less satisfactory match between the anomaly patterns and the GRTS. The second possibility is not indicated because the generally smooth vertical variation of the isotopic ages with stratigraphic position indicates that the isotopic ages represent eruption ages, close to the age of sediment deposition. Thus we suggest that either the third or fourth explanation is best.

Calibration of the Cenozoic portion of the GRTS by Berggren et al. (1985) is based on six selected radiometric ages distributed throughout the Cenozoic. It is possible that interlaboratory differences in standardization could account for the discrepancy in age between our preferred correlation and BKFV 85. For example, Samson and Alexander (1987) document interlaboratory reproducibility for measurement of K and Ar of only 1 to 2%.

An adjustment of the sea-floor spreading rates used to derive BKFV85 could also account for the differences. Berggren et al. (1985) determined a history of sea-floor spreading rates by dividing LaBrecque et al.'s (1977) sea floor magnetic anomaly pattern into three segments and by dating that anomaly pattern with isotopic ages at six tie points. Each segment represents a period of time during which the sea-floor spreading rate is taken to be constant. These periods are delimited by particular ages (inflection points) when the spreading rates must have changed. One such inflection point was placed at the older boundary of Anomaly 5 at 10.42 Ma. Slight errors in the ages of the tie points (in the late Neogene and in the Oligocene and Eocene) could change the age (inflection point) by as much as several million years; hence, its age is poorly constrained. We have recalibrated the GRTS, assuming that the inferred change in spreading occurred prior to 13 Ma rather than at 10.42 Ma. To do this, we extend the sea floor spreading rate calculated by Berggren et al. (1985) for the period younger than 10.42 Ma, back to ~13 Ma. Revised magnetic reversal ages for the preferred correlation defined by this calibration are listed under correlation (I*) in table 6. Using this new GRTS time scale, we have recalculated GRTS ages and Δ values (table 7). This results in a much better agreement between the isotopic and magnetic ages reflected by a reduction by half in the sum of squares of Δ.

The increase in age for this portion of the time scale partially closes the gap between magnetostratigraphic calibrations based on K-Ar isotopic ages obtained from mafic lava flows in Iceland (McDougall et al. 1984) and our earlier results for the Ngorora Formation for contemporaneous strata (Tauxe et al. 1985). The Icelandic calibration averages ~0.4 m.y. (3%) older than the new Ngorora results for the same GRTS boundaries, compared to ~0.8 m.y. (7%) older than Tauxe et al. (1985) (see fig. 8 for average ages of GRTS magnetozone boundaries derived from the new $^{40}Ar/^{39}Ar$ chronostratigraphy). The remaining discrepancy may be due to errors in

This content downloaded from 132.174.250.101 on March 28, 2019 15:13:49 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

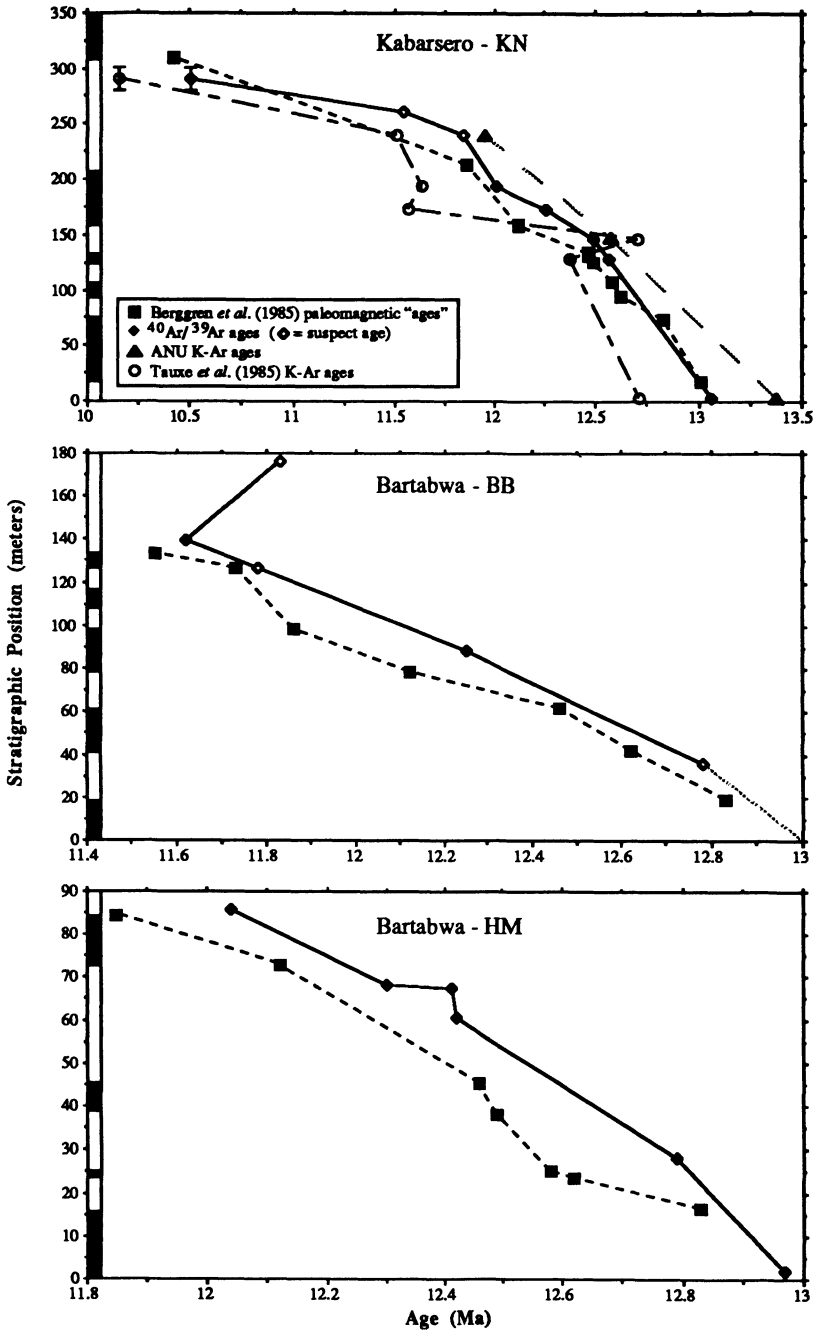


FIG. 9.—Plot of isotopic and paleomagnetic reversal ages *versus* stratigraphic height above base of section. $^{40}Ar/^{39}Ar$ ages indicated as "suspect" do not contain a clearly identifiable primary age component; the ages used here are the youngest single-crystal determinations (see table 1). Magnetostratigraphy appears along the left. (*a*) Kabarsero (KN), with conventional K-Ar dates of Tauxe et al. (1985) and new conventional K-Ar dates provided by I. McDougall. (*b*) Bartabwa main section (BB). The dashed extrapolation of the radioisotopic curve to the base of the section is a maximum age assuming that virtually all of anomaly 5AA is recorded. (*c*) Bartabwa hominoid section (HM).

This content downloaded from 132.174.250.101 on March 28, 2019 15:13:49 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

TABLE 8
COMPARISON OF CONVENTIONAL K-AR AND SINGLE-CRYSTAL $^{40}AR/^{39}AR$ AGES

Sample	K-Ar		$^{40}Ar/^{39}Ar$	
	Tauxe *et al.* (1985)	ANU (Table 4)	Eruption Age (Table 1)	'Bulk' Age
Samples used in interlaboratory comparison:				
RD-21	12.71 ± 0.09	13.38 ± 0.15	13.06 ± 0.03	13.06 ± 0.03
RD-51	12.70 ± 0.38	12.57 ± 0.13	12.49 ± 0.06	12.59 ± 0.49
RD-52	11.51 ± 0.22	11.96 ± 0.14	11.84 ± 0.07	12.44 ± 0.65
Average	12.31	12.64	12.46	12.70
Other Tauxe et al. (1985) samples:				
RD-54	12.37 ± 0.10	—	12.56 ± 0.04	12.56 ± 0.04
RD-53	11.57 ± 0.14	—	12.26 ± 0.07	12.27 ± 0.10
RD-55	11.63 ± 0.11	—	12.01 ± 0.03	12.01 ± 0.03
RD-23	10.16 ± 0.38	—	10.51 ± 0.03	10.74 ± 0.70

NOTE.—The 'bulk' $^{40}Ar/^{39}Ar$ age refers to a mean sample age encompassing all single-crystal determinations, including inferred contaminants.

correlation of magnetozone boundaries, to systematic differences in the dates from the two laboratories, or to unrecognized geologic problems.

Comparison of $^{40}Ar/^{39}Ar$ Determinations with Prior Conventional Dates.—The partial reconciliation of the Icelandic and East African calibration schemes is due primarily to improvement in the isotopic ages obtained for the Ngorora Formation. Tauxe et al. (1985) reported conventional K-Ar ages for seven K-feldspar samples from the Kabarsero section, all of which were redated herein by the single-crystal $^{40}Ar/^{39}Ar$ process. Six of these yielded older ages, with an overall average increase of ~0.3 m.y. (2–3%). These new $^{40}Ar/^{39}Ar$ dates also yield a much more plausible stratigraphic age chronology than that defined earlier by the K-Ar age assignments. We believe that the conventional K-Ar results reported by Tauxe et al. (1985) are suspect, possibly due to incomplete extraction of $^{40}Ar^*$ from the feldspar melt during fusion, but more likely due Ar gettering on the walls of the extraction vessel. We believe this process occurs as an alkali mirror is formed on the bottle walls during the higher temperature phases of RF fusion. Gettering of $^{40}Ar^*$ apparently occurs without the visible electrical discharges noted by Baksi (1974) to accompany Ar loss. In either case, artificially young K-Ar ages would result. One sample (RD-51), however, yielded a K-Ar age older than the $^{40}Ar/^{39}Ar$ age, most likely reflecting the presence of older contaminants now known to exist in the sample (see fig. 2*a*).

For the purposes of interlaboratory comparison, I. McDougall (ANU; written comm. 1988) has obtained conventional bulk-fusion K-Ar results on splits of three of the K-feldspar concentrates for which dates were reported in Tauxe et al. (1985) (table 8). One of the three samples (RD-21) in fact demonstrates a significant difference in age between the two laboratories at a 95% confidence level. The average age of the ANU determinations is 0.33 m.y. (3%) older than the Tauxe et al. (1985) ages, while the new $^{40}Ar/^{39}Ar$ determinations fall about midway between the two. The ANU age determinations are all older than the inferred $^{40}Ar/^{39}Ar$ eruption ages of these samples by an average of 0.18 m.y. However, when the ANU K-Ar results are compared to "bulk" $^{40}Ar/^{39}Ar$ ages computed from *all* analyses, including inferred contaminants, the difference is reduced to 0.06 m.y. (0.5%). Thus the bulk K-Ar ANU determinations tend to confirm the observation based on the new $^{40}Ar/^{39}Ar$ ages that the Tauxe et al. (1985) dates are too young by a few percent; on the other hand, the ANU ages are older than the $^{40}Ar/^{39}Ar$ results, apparently due to the incorporation of anomalously old feldspars.

Timing of Volcanic Activity.—In figure 10 we have plotted an age histogram of all of the Ngorora Formation and Tiim Phonolites feldspar $^{40}Ar/^{39}Ar$ analyses. None of the feld-

This content downloaded from 132.174.250.101 on March 28, 2019 15:13:49 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

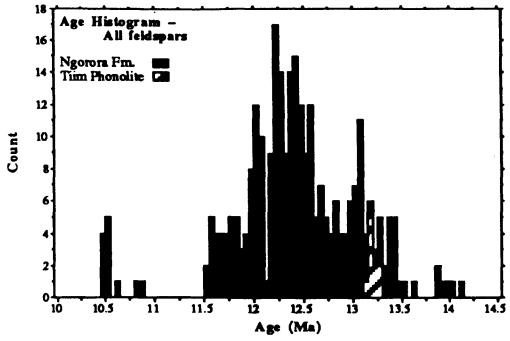


FIG. 10.—Cumulative histogram of single-crystal $^{40}Ar/^{39}Ar$ dates of the Ngorora Formation and underlying Tiim Phonolites Formation.

spars from the Ngorora Formation were older than about 14.2 Ma, indicating that detrital feldspar grains (for example, in feldspathic sandstone RD-52, fig. 2) were apparently derived from erosion of the products of earlier Ngorora-age volcanic activity, or from the upper Tiim Phonolites or related tephra. Older rock units in the region, such as the lower Tiim Phonolites or the Sidekh Phonolites (~15–16 Ma; Chapman and Brooks 1978), apparently had not yet been exposed by faulting within the Baringo Basin.

The more precise laser-fusion $^{40}Ar/^{39}Ar$ ages permit us to revise the interpretation in Tauxe et al. (1985) that the seven dated horizons in the Kabarsero section contained volcanic materials reworked from only three eruptive episodes. The overall distribution of our $^{40}Ar/^{39}Ar$ ages (fig. 10) does suggest in fact that volcanism clustered within three main eruptive sequences (~13 Ma, 12.5 to 11.5 Ma, and ~10.5 Ma). On a finer scale, the single-crystal $^{40}Ar/^{39}Ar$ data demonstrate that each successive dated stratum is measurably younger than the underlying dated strata, and therefore most likely defines a unique eruptive event. Where the age differences are small, as between volcaniclastic units represented by samples RD-54, RD-51, and RD-53, the distinct Ca/K ratios of the feldspars indicate a derivation from separate volcanic events.

The $^{40}Ar/^{39}Ar$ ages of the Ngorora Formation and the K-Ar ages of the overlying Ewalel Phonolites indicate that a widespread disconformity may be present between the two formations. Extrapolation of the age *vs.* stratigraphic position curves (fig. 9) indicates that the top of the Ngorora section at Bartabwa (BB) is about 11.2 Ma while at Kabarsero it is about 8 to 9 Ma. The top of the HM section at Bartabwa is truncated by faulting, but the youngest strata exposed are about 12 Ma old. Unfortunately, the extrapolated ages at Kabarsero are particularly ill-defined, because the uppermost ~130 m is undated, and of that the top 60 m is unexposed. The unconformity may represent as much as 4 m.y. at Bartabwa (BB) but more likely less time at Kabarsero.

Structural and Geomorphic Implications.—These age data have important implications for the structural and geomorphic evolution of the Baringo Basin during the middle Miocene. The voluminous phonolitic flows (Tiim Phonolites and stratigraphic equivalents) that underlie the Ngorora Formation completely filled and overflowed the middle Miocene Kenya Rift Valley and apparently formed an extensive tableland within the central part of Kenya (Lippard 1973). The mean $^{40}Ar/^{39}Ar$ age of anorthoclase phenocrysts from the uppermost Tiim flow at Bartabwa (BB) is 13.19 ± 0.04 Ma; at Kabarsero a whole-rock $^{40}Ar/^{39}Ar$ isochron age from the uppermost, aphyric Tiim Phonolite yields 13.15 ± 0.13 Ma. These ages indicate that phonolitic activity ended in this area about 13.2 Ma. Ngorora sedimentation began within 0.1 to 0.3 Ma of the cessation of phonolite activity, based on extrapolated ages for the base of the Ngorora Formation of 13.1 Ma at Kabarsero, 13.0 Ma at Bartabwa (HM), and ~12.9–13.1 Ma at Bartabwa (BB). In both the Kabarsero and Bartabwa areas, the oldest part of the sedimentary section is dominated by alluvial fan deposits and the youngest part by lacustrine deposits. Sediment accumulation rates estimated from the chronostratigraphies reflect this change, varying from high rates during the first 2 m.y. (~25 cm/1000 yr at Kabarsero and ~10 cm/1000 yr at Bartabwa), to lower rates during the last 1 m.y. (5 cm/1000 yr at Kabarsero). These lithologic characteristics and sedimentation rates imply that much local relief was present during early Ngorora times but became considerably more subdued by about 12 Ma. We interpret this change in sedimentation rate to indicate that the Ngorora sediments were deposited in a basin created

This content downloaded from 132.174.250.101 on March 28, 2019 15:13:49 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

within 0.1 to 0.3 m.y. of Tiim Phonolites volcanism by subsidence and that subsequently subsidence either did not occur or proceeded at a low rate. The Baringo Basin during Ngorora times may not have been a rift valley dominated by extensional tectonics, therefore, but a subsidence feature formed as a consequence of the rapid withdrawal from the lithosphere of large volumes of phonolitic magma.

Age of Hominoid.—Estimating with some confidence the ages of strata in which hominoid fossils have been found is an important focus of our efforts. A hominoid fourth, lower premolar KNM-BN 10489 (Hill et al. 1985; Hill and Ward 1988), tentatively assigned to *Proconsul* sp. indet., was found at 60.4 m above the base of the Bartabwa HM section, and 0.4 m below a lapilli tuff (ALD85-36) that yielded a primary volcanic age of 12.42 ± 0.03 Ma. Assuming constant sedimentation rates for this part of the section, the 0.4 m between the *Proconsul* and dated horizons represents less than 0.01 m.y. This premolar may then be the last known representative of *Proconsul* in the fossil record (Hill and Ward 1988).

ACKNOWLEDGMENTS.—We are pleased to acknowledge participants in the Baringo Basin field program: A. Hill, D. Pilbeam, A. K. Behrensmeyer, S. Ward, B. Brown, L. Jacobs, N. M. Johnson, B. F. Jacobs, and H. Staudigel. I. McDougall graciously provided analytical data for several conventional K-Ar determinations on Ngorora alkali feldspars. D. Clark provided invaluable assistance in the paleomagnetic laboratory. Thanks are also due to Niels Kylstra for assistance with the bootstrap statistics. This work was supported in part by NSF Grants BNS-8313340 and BNS-8616256 to G. Curtis and R. Drake, and OCE-88-11821 to L. Tauxe. Partial funding for equipment and operating expenses of the Geochronology Center of the Institute of Human Origins were provided by grants from the Getty Foundation and the Koch Foundation. Funds for equipment for the Scripps Paleomagnetic Laboratory were kindly provided by the W. M. Keck Foundation.

REFERENCES CITED

BAKSI, A. K., 1974, Spurious K-Ar ages due to radio-frequency induced electrical discharges in argon extraction systems: Can. Jour. Earth Sci., v. 11, p. 1020–1023.

BERGGREN, W. A.; KENT, D. V.; FLYNN, J.; and VAN COUVERING, J. A., 1985, Cenozoic geochronology: Geol. Soc. America Bull., v. 96, p. 1407–1418.

BISHOP, W. W., and CHAPMAN, G. R., 1970, Early Pliocene sediments and fossils from the northern Kenya Rift Valley: Nature, v. 226, p. 914–918.

———; ———; HILL, A.; and MILLER, J. A., 1971, Succession of Cainozoic vertebrate assemblages from the northern Kenya Rift Valley: Nature, v. 233, p. 389–394.

———, and PICKFORD, M. H. L., 1975, Geology, fauna, and palaeoenvironments of the Ngorora Formation, Kenya Rift Valley: Nature, v. 254, p. 185–192.

CEBULA, G. T.; KUNK, M. J.; MEHNERT, H. H.; NAESER, C. W.; OBRADOVICH, J. D.; and SUTTER, J. F., 1986, The Fish Canyon Tuff, a potential standard for the ^{40}Ar-^{39}Ar and Fission-track methods, *in* Sixth Int. Conf. Geochron., Cosmochron. Isotope Geol. (Cambridge, 1986) Abs.: TERRA Cognita, v. 6, n. 2, p. 139–140.

CHAPMAN, G. R., and BROOK, M., 1978, Chronostratigraphy of the Baringo Basin, Kenya Rift Valley, *in* BISHOP, W. W., ed., Geological Background to Fossil Man: Edinburgh, Scottish Academic Press, p. 207–223.

DEINO, A. L., and POTTS, R., 1990, Single-crystal $^{40}Ar/^{39}Ar$ dating of the Olorgesailie Formation, Southern Kenya Rift: Jour. Geophys. Res., in press.

DRAKE, R. E.; DEINO, A. L.; CURTIS, G. H.; SWISHER, C. S.; TURRIN, B.; MCCRORY, M.; and BECKER, T., 1988, Applications of $^{40}Ar/^{39}Ar$ dating of single-crystals by laser fusion: EOS, v. 69, p. 1502.

EFRON, B., 1982, The jackknife, the bootstrap and other resampling plans: Soc. Ind. Appl. Math, CBMC-Natl. Sci. Found. Mon., 99 p.

FISHER, R. A., 1953, Dispersion on a sphere: Proc. Royal Soc., v. 217, p. 295–305.

FLECK, R. J.; SUTTER, J. F.; and ELLIOT, D. H., 1977, Interpretation of discordant $^{40}Ar/^{39}Ar$ age-spectra of Mesozoic tholeiites from Antarctica: Geochim. Cosmochim. Acta, v. 41, p. 15–32.

HILL, A., 1989, Kipsaramon: a Miocene hominoid site in Kenya: Jour. Phys. Anthro. (abs.), v. 78, n. 2, p. 241.

———; DRAKE, R.; TAUXE, L.; MONAGHAN, M.; BARRY, J. C.; BEHRENSMEYER, A. K.; CURTIS, G.; FINE JACOBS, B.; JACOBS, L.; JOHNSON, N.; and PILBEAM, D., 1985, Neogene palaeontology and geochronology of the Baringo Basin, Kenya: Jour. Human Evol., v. 14, p. 759–773.

———, and WARD, S., 1988, Origin of the Hominidae: the record of African large hominoid evolution between 14 My and 4 My: Yearbook Phys. Anthro., v. 31, p. 49–83.

JOHNSON, N. M.; OPDYKE, N. D.; and LINDSAY, E. H., 1975, Magnetic polarity stratigraphy of

This content downloaded from 132.174.250.101 on March 28, 2019 15:13:49 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

Plio-Pleistocene terrestrial deposits and vertebrate faunas, San Pedro Valley, Arizona: Geol. Soc. America Bull., v. 86, p. 5–12.

KING, B. C., and CHAPMAN, G. R., 1972, Volcanism of the Kenya Rift Valley: Royal. Soc. (London) Philos. Trans.; v. 271 A, p. 185–204.

KYLSTRA, N., and TAUXE, L., 1989, Confidence regions for paleomagnetic data; a bootstrap approach: EOS (Trans. Am. Geophys. Union), v. 70, p. 1072.

LABRECQUE, J. L.; KENT, D. V.; and CANDE, S. C., 1977, Revised magnetic polarity time scale for Late Cretaceous and Cenozoic time: Geology, v. 5, p. 330–335.

LANPHERE, M. A., and DALRYMPLE, G. B., 1978, The use of $^{40}Ar/^{39}Ar$ data in evaluation of disturbed K-Ar systems, *in* ZARTMAN, R. E., ed., Short papers on the Fourth International Conference on Geochemistry, Cosmochemistry, and Isotope Geology: U.S. Geol. Survey Open File Rept. 78-701, p. 241–243.

LIPPARD, S. J., 1973, Plateau phonolite lava flows, Kenya: Geol. Mag. v. 110, p. 543–549.

LOBELLO, PH.; FÉRAUD, G.; HALL, C. M.; YORK, D.; LAVINA, P.; and BERNAT, M., 1987, $^{40}Ar/^{39}Ar$ step heating and laser fusion dating of a Quarternary volcanic from Neschers, Massif Central, France: the defeat of xenocrystic contamination: Chem. Geol., v. 66, p. 61–71.

MANDEL, J., 1964, The Statistical Analysis of Experimental Data, New York, Dover Publications, 410 p.

MAHOOD, G. A., and DRAKE, R. E., 1982, K-Ar dating young rhyolitic rocks: a case study of the Sierra La Primavera, Jalisco, Mexico: Geol. Soc. America Bull., v. 93, p. 1232–1241.

MCDOUGALL, I.; KRISTJANSSON, L.; and SAEMUNDSSON, K., 1984, Magnetostratigraphy and geochronology of northwest Iceland: Jour. Geophys. Res., v. 89, p. 7029–7060.

———, and HARRISON, T. M., 1988, Geochronology and Thermochronology by the $^{40}Ar/^{39}Ar$ Method: New York, Oxford University Press, 212 p.

MCINTYRE, G. A.; BROOKS, C.; COMPSTON, W.; and TUREK, A., 1966, The statistical assessment of Rb-Sr isochrons: Jour. Geophys. Res., v. 71, p. 5459–5468.

PICKFORD, M. H. L., 1975, Late Miocene sediments and fossils from the northern Kenya Rift Valley: Nature, v. 256, p. 279–284.

———, 1978, Geology, palaeoenvironments and vertebrate faunas of the mid-Miocene Ngorora Formation, Kenya, *in* BISHOP, W. W., ed., Geological Background to Fossil Man: Edinburgh, Scottish Academic Press, p. 237–262.

SAMSON, S. D., and ALEXANDER, E. C., JR., 1987, Calibration of the interlaboratory $^{40}Ar/^{39}Ar$ dating standard, MMhb-1: Chem. Geol. Iso. Geosci., v. 66, p. 27–34.

TAUXE, L.; MONAGHAN, M.; DRAKE, R.; CURTIS, G.; and STAUDIGEL, H., 1985, Paleomagnetism of Miocene East African rift sediments and the calibration of the geomagnetic reversal time scale: Jour. Geophys. Res., v. 90, p. 4639–4646.

TAYLOR, J. R., 1982, An Introduction to Error Analysis: Mill Valley, CA, Univ. Sci. Books, 270 p.

WATSON, G. S., 1956, A test for randomness: Mon. Not. Royal Astr. Soc., Geophys. Supp., v. 7, p. 160–161.

YORK, D., 1969, Least squares fitting of a straight line with correlated errors: Earth Planet. Sci. Lett., v. 5, p. 320–324.

———; HALL, C. M.; YANASE, Y.; HANES, J. A.; and KENYON, J., 1981, $^{40}Ar/^{39}Ar$ dating of terrestrial minerals with a continuous laser: Geophys. Res. Lett., v. 8, p. 1136–1138.

This content downloaded from 132.174.250.101 on March 28, 2019 15:13:49 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).